BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The nerve plexuses and the autonomic nervous system

The written notes for the nerve plexuses and autonomic nervous system lectures: the five spinal nerve plexuses and the regions they serve, the somatic and autonomic motor pathways, and the two autonomic divisions with their ganglia and plexuses. Module 5 competencies 191 to 193, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 191. Nerve plexuses

In your course notes: Module 5 packet, Chapter 12, competency 191 (page 83).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Nerve plexuses and injuries.

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

There are 31 pairs of spinal nerves: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal. The anterior rami of most of them do not run straight to their targets. Instead the anterior rami of neighboring spinal nerves join, swap axons, and branch again to form a network called a nerve plexus. Inside the plexus the axons are sorted and regrouped, and they leave in named nerves. Because of this regrouping, each nerve that leaves a plexus carries axons from several spinal nerves, and each body region gets axons from more than one spinal nerve. Damage to one spinal nerve weakens a region but does not usually wipe out all function there.

The thoracic region is the exception. The anterior rami of T2 to T12 do not form plexuses. They run straight out between the ribs as the intercostal nerves (T12 runs below the last rib, so it is called the subcostal nerve). They supply the intercostal muscles, the muscles of the abdominal wall, and the skin of the chest and abdomen. T1 is the one thoracic nerve that joins a plexus: most of it goes into the brachial plexus.

The five plexuses at a glance

Plexus	Formed by (anterior rami)	Main nerves	Region served
Cervical plexus	C1 to C4, with a small contribution from C5	Phrenic nerve (C3 to C5) and small nerves to the neck	The skin and muscles of the neck and upper shoulder; the phrenic nerve is the motor supply to the diaphragm
Brachial plexus	C5 to C8 and T1	Axillary, musculocutaneous, radial, median, and ulnar nerves	The shoulder and the entire upper limb
Lumbar plexus	L1 to L4	Femoral and obturator nerves	The lower abdominal wall, the external genitals, and the front and inner (medial) thigh
Sacral plexus	L4 to L5 and S1 to S4	Sciatic nerve (the largest nerve in the body), gluteal nerves, and pudendal nerve	The buttock, the perineum, the back of the thigh, and most of the leg and foot
Coccygeal plexus	S4, S5, and the coccygeal nerve (Co1)	Small nerves to the skin	A small area of skin over the coccyx

- Memory tip: the plexuses run head to tail in the same order as the cord regions that build them. Cervical for the neck and diaphragm, brachial for the upper limb, lumbar for the abdominal wall and front of the thigh, sacral for the buttock and the rest of the lower limb, and coccygeal for the skin over the coccyx.
- The lumbar and sacral plexuses share L4, and the sacral and coccygeal plexuses share S4. That is why the lower limb is sometimes described as being served by one lumbosacral plexus.
- The brachial plexus takes in the first thoracic nerve. When you bracket it on a drawing, make sure the bracket reaches T1.

Cervical plexus (C1 to C4, with a little of C5)

The cervical plexus lies deep in the side of the neck, under the sternocleidomastoid muscle. Its branches are of two kinds. The cutaneous branches come out around the back edge of the sternocleidomastoid and spread to the skin of the head, neck, and upper shoulder. The deep branches run to the muscles of the neck and, through the phrenic nerve, to the diaphragm.

Nerve	Spinal nerves	Where it goes
Lesser occipital nerve	C2	Skin of the scalp behind and above the ear
Great auricular nerve	C2, C3	Skin over the ear and the angle of the jaw
Transverse cervical nerve	C2, C3	Skin of the front and side of the neck
Supraclavicular nerves	C3, C4	Skin of the upper shoulder and the upper chest, above and below the clavicle
Ansa cervicalis	C1 to C3	The strap muscles in the front of the neck below the hyoid
Phrenic nerve	C3 to C5	The diaphragm. It runs down through the thorax, passing in front of the root of the lung, and is the only motor supply to the diaphragm
Segmental branches	C1 to C4	The deep muscles of the neck, with help to the sternocleidomastoid and trapezius

- The phrenic nerve is the branch that matters most. C3, 4, 5 keeps the diaphragm alive: an injury to the cord at or above that level can stop breathing.
- The phrenic nerve comes from the cervical plexus, not the brachial plexus. When you draw it, start it in the neck above the brachial plexus.

Brachial plexus (C5 to T1)

The brachial plexus starts in the lower neck, passes behind the clavicle, and runs into the axilla (armpit). It is the largest and most rearranged of the plexuses. The five anterior rami (the roots) join into three trunks (superior, middle, inferior). Each trunk splits into an anterior and a posterior division. The divisions regroup into three cords (lateral, posterior, medial), named for their position around the axillary artery. The cords give off the five main nerves of the upper limb. Roots, trunks, divisions, cords, branches: Really Tired, Drink Coffee, Be awake.

Nerve	Spinal nerves (approximate)	Where it goes
Axillary nerve	C5, C6	The shoulder: the muscles that cap the shoulder joint and the skin over the top of the shoulder
Musculocutaneous nerve	C5 to C7	The muscles on the front of the arm and the skin of the lateral (thumb side) forearm
Radial nerve	C5 to T1	The back of the whole limb: the muscles on the back of the arm and forearm and the skin of the back of the arm, forearm, and hand
Median nerve	C5 to T1	Most of the muscles on the front of the forearm, some muscles of the thumb side of the hand, and the skin of the lateral palm and the thumb side fingers
Ulnar nerve	C8, T1	Some muscles on the front of the forearm, most of the small muscles of the hand, and the skin of the medial (little finger side) hand

- Smaller branches leave the roots, trunks, and cords before the five main nerves form. They supply the muscles that move the scapula and the muscles of the chest wall and back that attach to the arm, and the skin of the medial arm and forearm. Books list the exact spinal nerves for each branch a little differently, so learn the range for the whole plexus (C5 to T1) and the main nerve names.
- Where to find them on a model: the axillary nerve wraps around the back of the top of the humerus. The musculocutaneous nerve runs down the front of the arm. The radial nerve spirals around the back of the humerus. The median nerve runs down the middle of the front of the arm and forearm and through the carpal tunnel at the wrist. The ulnar nerve passes behind the medial epicondyle of the humerus, the funny bone spot, and then down the medial forearm.

Lumbar plexus (L1 to L4)

The lumbar plexus forms inside the psoas major muscle, beside the lumbar vertebrae. Its branches run to the lower abdominal wall, the external genitals, and the front and inner thigh. The two large nerves are the femoral and obturator nerves.

Nerve	Spinal nerves	Where it goes
Iliohypogastric nerve	L1	The muscles and skin of the lower abdominal wall and the skin of the upper buttock
Ilioinguinal nerve	L1	The lower abdominal wall, the skin of the upper inner thigh, and the skin of the external genitals
Genitofemoral nerve	L1, L2	The skin of the external genitals and the skin of the upper front thigh
Lateral femoral cutaneous nerve	L2, L3	The skin of the outer (lateral) thigh
Femoral nerve	L2 to L4	The muscles on the front of the thigh and the skin of the front and inner thigh; its long cutaneous branch, the saphenous nerve, supplies the skin of the inner leg and foot
Obturator nerve	L2 to L4	The muscles on the inner thigh and the skin of the inner thigh. It leaves the pelvis through the obturator foramen

- The femoral nerve passes under the inguinal ligament into the front of the thigh. The obturator nerve goes through the obturator foramen of the hip bone to the inner thigh. The names tell you the route.

Sacral plexus (L4 to S4)

The sacral plexus lies on the back wall of the pelvis, in front of the sacrum and the piriformis muscle. It serves the buttock, the perineum, the back of the thigh, and almost all of the leg and foot. Its largest branch, the sciatic nerve, is the largest and longest nerve in the body.

Nerve	Spinal nerves	Where it goes
Superior gluteal nerve	L4 to S1	The muscles on the side of the buttock that steady the pelvis when you walk
Inferior gluteal nerve	L5 to S2	The large muscle of the buttock
Posterior femoral cutaneous nerve	S1 to S3	The skin of the lower buttock and the back of the thigh
Sciatic nerve	L4 to S3	Leaves the pelvis through the greater sciatic foramen, passes under the gluteus maximus, and runs down the back of the thigh to the muscles there. Above the knee it splits into the tibial and common fibular nerves
Tibial nerve (from the sciatic)	L4 to S3	The back of the leg and the sole of the foot, muscles and skin
Common fibular (peroneal) nerve (from the sciatic)	L4 to S2	The side and front of the leg and the top of the foot, muscles and skin. It wraps around the neck of the fibula just below the knee, where it sits close under the skin
Pudendal nerve	S2 to S4	The perineum: the muscles of the pelvic floor and the external anal and urethral sphincters, and the skin of the external genitals

- The sciatic nerve is really two nerves, the tibial and the common fibular, wrapped in one sheath. The tibial half serves the back of the leg and the sole; the fibular half serves the front and side of the leg and the top of the foot.
- The pudendal nerve is the nerve of the perineum. It carries sensation from the external genitals and controls the sphincters you consciously close.

Coccygeal plexus (S4, S5, and Co1)

The smallest plexus. It is formed by the anterior rami of S4 and S5 and the single coccygeal nerve, and it lies on the coccygeus muscle at the bottom of the pelvis. Its branches, the anococcygeal nerves, supply a small area of skin over the coccyx and around it.

Predicting the effect of plexus damage

Each plexus serves one region, so the plexus that is damaged predicts the region that loses movement and sensation. To answer a question, first name the plexus, then name the region it serves. The damage shows up on the same side of the body as the injured plexus.

Plexus damaged	Region affected
Cervical plexus	Weakness or numbness of the neck and shoulder; if the phrenic nerve is involved, the diaphragm on that side is weak and breathing is affected
Brachial plexus	Weakness and sensory loss across the shoulder and the whole upper limb on that side
Lumbar plexus	Weakness and sensory loss of the lower abdominal wall and the front and inner surfaces of the thigh
Sacral plexus	Weakness and sensory loss of the buttock, the back of the thigh, and the leg and foot
Coccygeal plexus	Numbness of the small patch of skin over the coccyx

Why the network matters

- The plexus design is a safety feature. Each limb region draws axons from several spinal nerves, so a single damaged spinal nerve root weakens an area but rarely paralyzes it completely.
- Damage to the plexus itself is different. It can damage axons from several spinal nerves at once, so it can take out a whole region. A brachial plexus injury affects the upper limb, and a lumbar or sacral plexus injury affects the lower limb.
- The phrenic nerve leaves the cervical plexus and is the only motor supply to the diaphragm. An injury to the cord at or above C3 to C5 can stop breathing, which is why a high neck injury is an emergency for the airway, not just for the limbs.

What this means for patients

- Damage to a plexus shows up as weakness and numbness in the region that plexus serves, on the same side as the injury. Name the plexus, then name the region.
- A cut or crushed phrenic nerve weakens the diaphragm on that side, so the patient's breathing suffers. If the cord is damaged at or above C3 to C5, both phrenic nerves lose their supply and the patient cannot breathe without help.
- Because a thoracic nerve does not join a plexus, damage to one intercostal nerve affects only one strip of chest or abdominal wall, not a whole region.

Competency 192. Autonomic two-neuron pathway

In your course notes: Module 5 packet, Chapter 13, competency 192 (page 85).

Dr. Rennie's video for this part: Autonomic nervous system concept video, chapter Autonomic nervous system overview, Somatic versus autonomic basics, and The two-neuron pathway concept.

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

The motor side of the peripheral nervous system has two parts. The somatic nervous system controls skeletal muscle, and that control is usually voluntary. The autonomic nervous system (ANS) controls cardiac muscle, smooth muscle, and glands, and that control is involuntary. Both start in the central nervous system. The big structural difference is how many neurons it takes to get from the CNS to the effector: one for somatic, two for autonomic.

Somatic and autonomic motor systems compared

Feature	Somatic nervous system	Autonomic nervous system
Effectors	Skeletal muscle	Cardiac muscle, smooth muscle, and glands
Control	Voluntary	Involuntary
Motor neurons from CNS to effector	One	Two, with a synapse in an autonomic ganglion
Where the motor cell bodies are	Anterior horn of the cord or a motor nucleus of the brainstem	First neuron in the brainstem or the cord; second neuron in an autonomic ganglion
Myelin	The axon is myelinated	The first axon is lightly myelinated; the second axon is unmyelinated
Effect on the effector	Always excites; causes contraction	Can excite or inhibit
Awareness	Normally conscious	Normally not conscious

- The somatic system only ever excites its muscle. If a skeletal muscle needs to relax, its motor neuron just stops firing. Inhibition belongs in the autonomic column: the ANS can slow the heart as well as speed it up.
- The somatic pathway is one myelinated axon from a cell body in the anterior horn of the spinal cord (or a brainstem motor nucleus) straight to the skeletal muscle. That is the whole pathway.

The two-neuron pathway, in order

The autonomic system always uses two motor neurons in a row, and they meet in a ganglion. Trace the signal from the CNS to the effector.

1. **Preganglionic neuron.** Its cell body is in the CNS, in a brainstem nucleus or in the lateral gray matter of the spinal cord. Its axon is thin and lightly myelinated. It leaves the CNS in a cranial nerve or an anterior root and ends in an autonomic ganglion.
2. **Autonomic ganglion.** A cluster of neuron cell bodies outside the CNS. Here the preganglionic axon synapses with the second neuron. This is the only synapse on the way out.
3. **Postganglionic neuron.** Its cell body sits in the ganglion. Its unmyelinated axon runs from the ganglion out to the effector and carries the signal the rest of the way.
4. **Effector.** Cardiac muscle, smooth muscle, or a gland receives the signal.

Preganglionic neuron
The first neuron. Cell body in the brainstem or the lateral gray matter of the cord; lightly myelinated axon that leaves the CNS and ends in a ganglion.

Autonomic ganglion
A cluster of motor cell bodies outside the CNS where the preganglionic axon synapses with the postganglionic neuron.

Postganglionic neuron
The second neuron. Cell body in the ganglion; unmyelinated axon that runs to the effector.

Effector
The tissue that responds: cardiac muscle, smooth muscle, or a gland.

- The preganglionic axon is the lightly myelinated one and the postganglionic axon is unmyelinated. Do not swap them on your drawing.
- Memory tip: pre means before the ganglion and post means after it.
- Do not mix up the two kinds of ganglia. A posterior root ganglion holds sensory cell bodies and has no synapses. An autonomic ganglion holds motor cell bodies and is where the synapse happens.
- The ANS also has sensory neurons. They carry signals from receptors in the organs and blood vessels back to the CNS, and those signals let the CNS adjust the organs. The two-neuron pathway is the motor side only.

Competency 193. Autonomic ganglia and divisions

In your course notes: Module 5 packet, Chapter 13, competency 193 (page 86).

Dr. Rennie's video for this part: Autonomic nervous system concept video, chapter Parasympathetic versus sympathetic, Parasympathetic routes and ganglia, and Sympathetic routes in the body.

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

The ANS has two divisions, the sympathetic and the parasympathetic. Most organs get axons from both, and the two usually have opposite effects. The two divisions differ in where their preganglionic neurons start in the CNS and in where their ganglia sit. Those two facts, origin and ganglion location, set the lengths of the axons and give the divisions their other names.

The sympathetic division

The sympathetic division is the fight or flight division. It takes over during exercise, excitement, and emergencies. It is also called the thoracolumbar division, because its preganglionic cell bodies sit in the lateral horns of the T1 to L2 spinal segments. Its ganglia lie close to the spinal cord, so its preganglionic axons are short and its postganglionic axons are long.

Sympathetic trunk

A chain of ganglia on each side of the vertebral column, running from the base of the skull to the coccyx. Axons link one ganglion to the next. The trunk connects to the spinal nerves through the white and gray rami communicantes.

Sympathetic trunk ganglia

The ganglia of the chain, also called paravertebral or chain ganglia. Postganglionic neurons here send their axons back into the spinal nerves to reach the skin, blood vessels, and the organs of the thorax.

Prevertebral ganglia

Also called collateral ganglia. They lie in front of the vertebral column, close to the large branches of the abdominal aorta. They are the celiac, superior mesenteric, inferior mesenteric, aorticorenal, and renal ganglia. Preganglionic axons reach them by passing through the sympathetic trunk without synapsing, in nerves called splanchnic nerves.

White ramus communicans

The short branch that carries the lightly myelinated preganglionic axon from the spinal nerve into the sympathetic trunk. White because of the myelin. Only present at T1 to L2, where the preganglionic neurons are.

Gray ramus communicans

The short branch that carries the unmyelinated postganglionic axon from the trunk ganglion back into the spinal nerve. Gray because there is no myelin. Present at every spinal nerve, since every spinal nerve carries sympathetic axons to skin and vessels.

Follow a sympathetic signal to a sweat gland in the skin

1. Lateral horn (T1 to L2): the preganglionic cell body sends out its axon.
2. Anterior root and spinal nerve: the preganglionic axon leaves the cord.
3. White ramus communicans: the preganglionic axon enters the sympathetic trunk.
4. Sympathetic trunk ganglion: the preganglionic axon synapses with the postganglionic neuron.
5. Gray ramus communicans: the postganglionic axon returns to the spinal nerve.
6. Effector: the axon runs out in a ramus of the spinal nerve to a sweat gland, an arrector pili muscle, or a blood vessel in the skin.

For the abdominal and pelvic organs, the path differs after the white ramus. The preganglionic axon passes through the trunk ganglion without synapsing and travels in a splanchnic nerve to a prevertebral ganglion. It synapses there. The postganglionic axon then follows the arteries to the organ.

- A preganglionic sympathetic axon that enters the trunk has three choices: synapse in the trunk ganglion at its own level, travel up or down the chain and synapse in a trunk ganglion at another level, or pass straight through and synapse in a prevertebral ganglion. That is how T1 to L2 can reach every part of the body, from the pupil to the pelvis.

The parasympathetic division

The parasympathetic division is the rest and digest division. It conserves energy and runs routine work such as digestion, urination, and defecation. It is also called

the craniosacral division. Its preganglionic cell bodies sit in the brainstem nuclei of four cranial nerves (CN III, VII, IX, and X) and in the lateral gray matter of the S2 to S4 spinal segments.

Terminal ganglia

The ganglia of the parasympathetic division. They lie close to the organ or inside its wall (intramural ganglia). So the preganglionic axons are long and the postganglionic axons are short.

Vagus nerve (CN X)

Carries preganglionic axons to terminal ganglia in the walls of the thoracic and abdominal organs. It carries most of the parasympathetic supply to the chest and abdomen, which is why one nerve can slow the heart, narrow the airways, and speed digestion.

Pelvic splanchnic nerves

From S2 to S4. They carry preganglionic axons to the pelvic organs and the last part of the large intestine.

The four named terminal ganglia of the head

Ganglion	Cranial nerve that brings the preganglionic axons	Target
Ciliary ganglion	CN III	The muscles that constrict the pupil and focus the lens
Pterygopalatine ganglion	CN VII	The lacrimal gland and the glands of the nasal cavity
Submandibular ganglion	CN VII	The submandibular and sublingual salivary glands
Otic ganglion	CN IX	The parotid salivary gland

- CN VII reaches two ganglia, one for tears and nasal mucus and one for the two salivary glands under the jaw. CN IX reaches the parotid. CN III reaches the eye. CN X skips the head and takes care of the chest and abdomen, synapsing in ganglia in the organ walls.

Sympathetic and parasympathetic divisions compared

Feature	Sympathetic	Parasympathetic
Other name	Thoracolumbar division	Craniosacral division
Origin in the CNS	Lateral horns of the T1 to L2 spinal segments	Brainstem nuclei of CN III, VII, IX, and X, and the S2 to S4 spinal segments
Location of ganglia	Close to the spinal cord: sympathetic trunk ganglia and prevertebral ganglia	Close to or in the wall of the organ: terminal ganglia
Named ganglia	Sympathetic trunk (chain) ganglia; prevertebral ganglia: celiac, superior mesenteric, inferior mesenteric, aorticorenal, renal	Terminal ganglia, including the ciliary, pterygopalatine, submandibular, and otic
Preganglionic axon	Short	Long
Postganglionic axon	Long	Short
Overall role	Fight or flight; prepares the body for action	Rest and digest; conserves energy

- Memory tip: the other two names map the origins. Thoracolumbar means the preganglionic neurons start from T1 to L2. Craniosacral means they start in the brainstem and in S2 to S4. The ganglia follow: sympathetic near the cord, parasympathetic at the organ.

- Sympathetic leaves the middle of the cord, T1 to L2. Parasympathetic leaves the two ends, the brainstem and S2 to S4. Thoracolumbar is sympathetic and craniosacral is parasympathetic; check that you did not swap the names.
- Ganglion location sets axon length. Ganglia near the cord mean a short first axon and a long second axon (sympathetic). Ganglia at the organ mean a long first axon and a short second axon (parasympathetic).

The autonomic plexuses

In the thorax, abdomen, and pelvis, autonomic axons form networks around the large blood vessels. These are the autonomic plexuses. They hold sympathetic and parasympathetic axons together, along with sensory axons from the organs. Some also hold sympathetic ganglia. For example, the celiac ganglia sit inside the celiac plexus.

Plexus	Where it is and what it supplies
Cardiac plexus	Near the base of the heart; supplies the heart
Pulmonary plexus	Around the bronchi at the root of each lung; supplies the bronchial tree
Celiac (solar) plexus	The largest; surrounds the celiac trunk and supplies most of the upper abdominal organs
Superior and inferior mesenteric plexuses	Around the mesenteric arteries; supply the small and large intestines
Renal plexus	Around the renal arteries; supplies the kidneys and ureters
Hypogastric plexus	In front of the lower lumbar vertebrae and in the pelvis; supplies the pelvic organs

- An autonomic plexus is a mixed network around a vessel. It is not the same thing as a ganglion, though a ganglion can sit inside one. And it is not the same thing as a spinal nerve plexus from competency 191, which is a regrouping of anterior rami for skin and skeletal muscle.

The enteric plexuses

The wall of the digestive tract has its own networks of neurons, the enteric plexuses. They run from the esophagus to the anus and form the enteric nervous system, which has about 100 million neurons. These networks can run the gut wall on their own, and the ANS adjusts their activity.

Myenteric plexus

Lies in the muscularis externa, between the circular and longitudinal muscle layers. It mainly controls the movement of the gut.

Submucosal plexus

Lies in the submucosa, between the muscularis mucosae and the circular muscle layer. It mainly controls the secretions of the gut.

- Myenteric means in the muscle, and it runs the muscle. Submucosal means under the mucosa, and it runs the glands of the mucosa.
- The enteric nervous system is independent enough to be called the brain of the gut. Its plexuses can keep the gut wall contracting in rhythm and its glands secreting even when the connections to the CNS are cut.

What this means for patients

- The two divisions usually pull an organ in opposite directions, so the body can adjust an organ by easing off one division as well as by pushing the other. Heart rate, for example, rises when parasympathetic input is withdrawn, even before the sympathetic division speeds the heart up.
- The vagus nerve carries most of the parasympathetic supply to the chest and abdomen. Its long reach is why one nerve can slow the heart, narrow the airways, and speed digestion, and why damage to it affects so many organs at once.
- Because the gut has its own nervous system, digestion keeps going even when the autonomic connections to it are cut. The ANS turns the enteric plexuses up or down; it does not run them.

Worked questions

Eight questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 191

Which plexus is formed by the anterior rami of L1 to L4?

A. Cervical plexus

B. Brachial plexus

C. Lumbar plexus

D. Sacral plexus

Answer: C. The lumbar plexus is formed by L1 to L4. Its main nerves are the femoral and obturator nerves, and it serves the lower abdominal wall, the external genitals, and the front and inner thigh.

Why the others are wrong

- **A** The cervical plexus is formed by C1 to C4 with a small part from C5. It serves the neck and upper shoulder and gives off the phrenic nerve.
- **B** The brachial plexus is formed by C5 to C8 and T1. It serves the shoulder and the whole upper limb.
- **D** The sacral plexus is formed by L4 to L5 and S1 to S4. It shares L4 with the lumbar plexus but is a separate plexus serving the buttock, perineum, back of the thigh, and the leg and foot.

QUESTION 2 · COMPETENCY 191

Which spinal nerves form the brachial plexus?

A. C5 to C8 and T1

B. C1 to C4

C. L4 to S4

D. C5 to C8 only

Answer: A. The brachial plexus is built from the anterior rami of C5, C6, C7, C8, and T1. It is the one plexus that takes in a thoracic nerve, so a bracket on a drawing has to reach T1.

Why the others are wrong

- **B** C1 to C4 form the cervical plexus, which serves the neck and upper shoulder.
- **C** L4 to S4 form the sacral plexus, which serves the buttock and most of the lower limb.
- **D** C5 to C8 are only part of the plexus. T1 joins them, and leaving it out is the most common mistake on the drawing.

QUESTION 3 · COMPETENCY 191

A patient has an injury to the right sacral plexus. Which region would you expect to be weak and numb?

A. The front and inner thigh

B. The neck and upper shoulder

C. The small patch of skin over the coccyx

D. The buttock, the back of the thigh, and the leg and foot on the right side

Answer: D. The sacral plexus serves the buttock, the perineum, the back of the thigh, and most of the leg and foot through the gluteal, pudendal, and sciatic nerves. Damage shows up in that region on the same side as the injury.

Why the others are wrong

- **A** The front and inner thigh are served by the lumbar plexus, through the femoral and obturator nerves.
- **B** The neck and upper shoulder are served by the cervical plexus.
- **C** The skin over the coccyx is served by the coccygeal plexus, which is formed by S4, S5, and the coccygeal nerve.

QUESTION 4 · COMPETENCY 191

The phrenic nerve, the motor supply to the diaphragm, comes from which plexus?

A. Brachial plexus

B. Cervical plexus

C. Lumbar plexus

D. None; it is a thoracic intercostal nerve

Answer: B. The phrenic nerve leaves the cervical plexus and is built from C3, C4, and C5. It runs down through the thorax to the diaphragm, and a cord injury at or above that level can stop breathing.

Why the others are wrong

- **A** The brachial plexus starts at C5 and serves the upper limb. The phrenic nerve starts higher, in the cervical plexus, and does not go to the limb.
- **C** The lumbar plexus is L1 to L4 and serves the lower abdominal wall and the front and inner thigh, nowhere near the diaphragm.
- **D** The intercostal nerves are the anterior rami of T2 to T12, which run between the ribs and do not form a plexus. They supply the intercostal muscles, not the diaphragm.

QUESTION 5 · COMPETENCY 192

Where is the cell body of a postganglionic autonomic neuron?

A. In an autonomic ganglion

B. In the lateral gray matter of the spinal cord

C. In a posterior root ganglion

D. In the anterior horn of the spinal cord

Answer: A. The postganglionic neuron is the second neuron of the two-neuron pathway. Its cell body sits in the autonomic ganglion, where the preganglionic axon synapsed with it, and its unmyelinated axon runs from the ganglion to the effector.

Why the others are wrong

- **B** The lateral gray matter of the cord holds the cell bodies of preganglionic neurons, the first neurons of the pathway.
- **C** A posterior root ganglion holds sensory cell bodies and has no synapses in it. It is not part of the motor pathway.
- **D** The anterior horn holds somatic motor neurons, the single neurons that run straight to skeletal muscle.

QUESTION 6 · COMPETENCY 192

Which feature belongs to the somatic motor system and not to the autonomic motor system?

A. Two motor neurons with a synapse in a ganglion

B. It can inhibit its effector as well as excite it

C. One myelinated axon runs from the CNS all the way to the effector

D. Its effectors are cardiac muscle, smooth muscle, and glands

Answer: C. The somatic pathway is a single neuron. Its cell body is in the anterior horn of the cord or a brainstem motor nucleus, and its myelinated axon runs without a break all the way to the skeletal muscle.

Why the others are wrong

- **A** Two neurons in a row with a synapse in an autonomic ganglion is the autonomic pattern.
- **B** Only the autonomic system can inhibit. The somatic system always excites its muscle; to relax a skeletal muscle, the motor neuron simply stops firing.
- **D** Cardiac muscle, smooth muscle, and glands are the autonomic effectors. The somatic effector is skeletal muscle.

QUESTION 7 · COMPETENCY 193

Where are the cell bodies of parasympathetic preganglionic neurons?

A. The lateral horns of the T1 to L2 spinal segments

B. The brainstem nuclei of CN III, VII, IX, and X, and the S2 to S4 spinal segments

C. The sympathetic trunk ganglia

D. The prevertebral ganglia

Answer: B. The parasympathetic division is the craniosacral division. Its preganglionic neurons start in the brainstem (in the nuclei of CN III, VII, IX, and X) and in the sacral cord at S2 to S4, the two ends of the CNS.

Why the others are wrong

- **A** The lateral horns of T1 to L2 hold the sympathetic preganglionic neurons. That is why the sympathetic division is called thoracolumbar.
- **C** The sympathetic trunk ganglia hold sympathetic postganglionic cell bodies, not preganglionic ones, and they belong to the other division.
- **D** The prevertebral ganglia (celiac, mesenteric, aorticorenal, renal) hold sympathetic postganglionic cell bodies in front of the vertebral column.

QUESTION 8 · COMPETENCY 193

Which plexus lies in the muscularis externa of the gut wall, between the circular and longitudinal muscle layers, and mainly controls gut movement?

A. Submucosal plexus

B. Celiac plexus

C. Hypogastric plexus

D. Myenteric plexus

Answer: D. The myenteric plexus is one of the two enteric plexuses. It sits between the two muscle layers of the muscularis externa and controls the movement of the gut. Myenteric means in the muscle.

Why the others are wrong

- **A** The submucosal plexus is the other enteric plexus. It lies in the submucosa, between the muscularis mucosae and the circular muscle layer, and mainly controls secretion.
- **B** The celiac plexus is an autonomic plexus outside the gut. It surrounds the celiac trunk and supplies most of the upper abdominal organs.
- **C** The hypogastric plexus is an autonomic plexus in front of the lower lumbar vertebrae and in the pelvis. It supplies the pelvic organs.